

Data-Driven Computer Animation

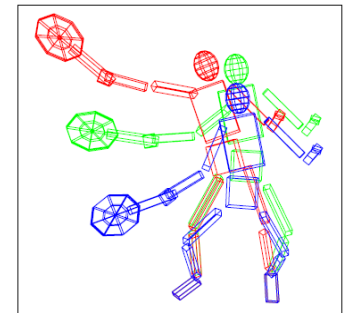
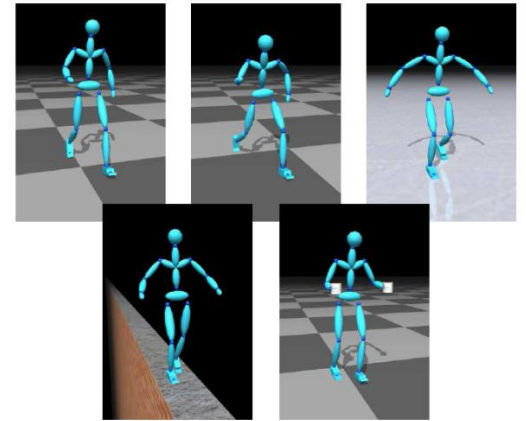
Lecture 5.

Motion Editing and Synthesis

Taku Komura

Overview

- Motion Editing
 - Motion Warping
 - Motion Blending
 - Dynamic Time Warping
- Motion databases



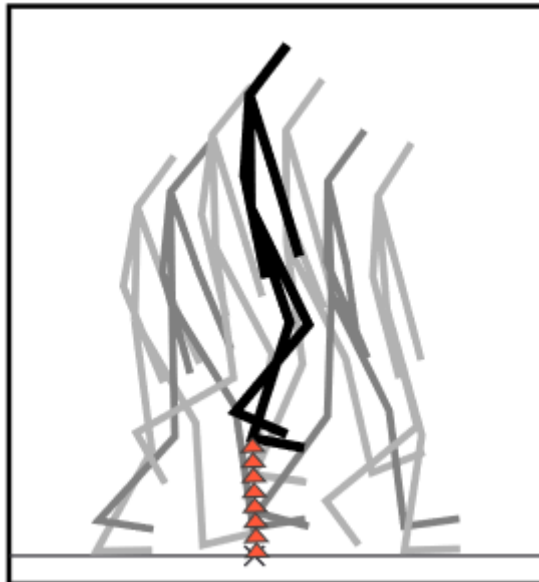
Motion Editing / Motion Blending : Background

- We learnt about motion capture
- It can provide realistic movements for characters
- However,...



Motion needs adjustments

- There might be obstacles in the scene – the character needs to avoid them



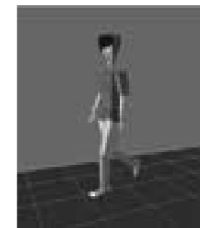
Motion needs adjustments

- There might be obstacles in the scene – the character needs to avoid them
- The character's body size is different from those of humans
 - The hand not reaching the right position
 - Need to “retarget” the motion to the character size while maintaining constraints



Motion needs adjustments

- There might be obstacles in the scene – the character needs to avoid them
- The character's body size is different from those of humans
 - The hand not reaching the right position
 - Need to “retarget” the motion to the character size while maintaining constraints
- The style of motion might not be good
 - Want to exaggerate some expressions



(a) Normal walk



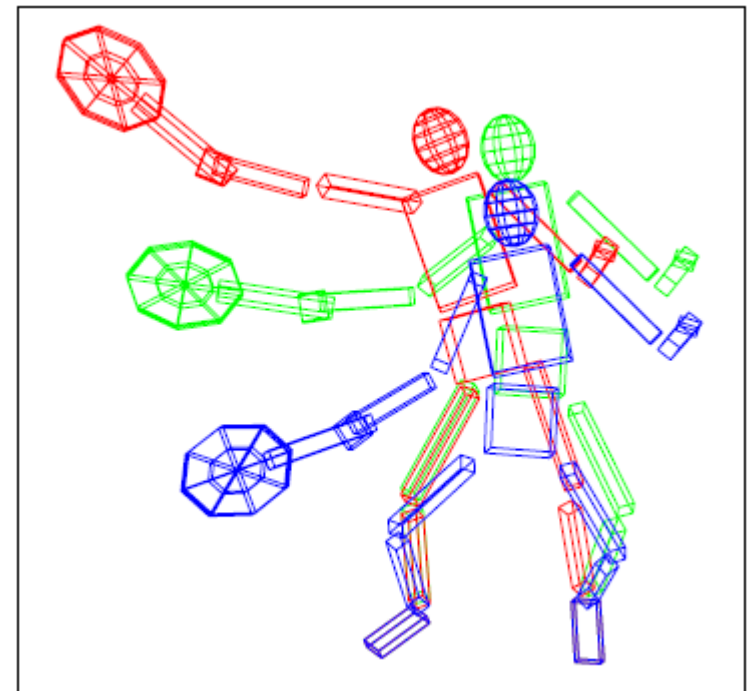
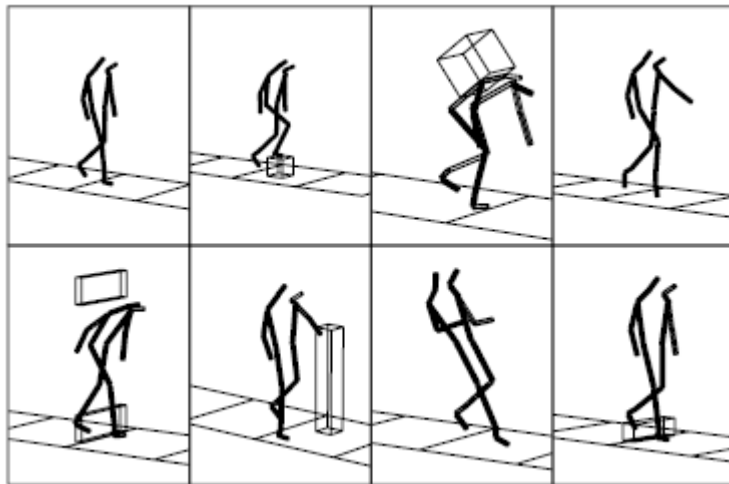
(b) Tired walk

Editing motions

- Need an efficient method to edit the trajectories of the body
- Whilst satisfying constraints
- Must be interactive – the tools are used by animators

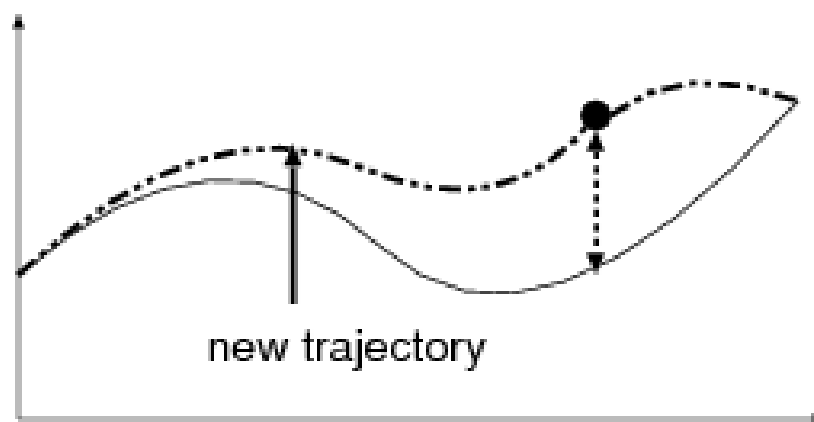
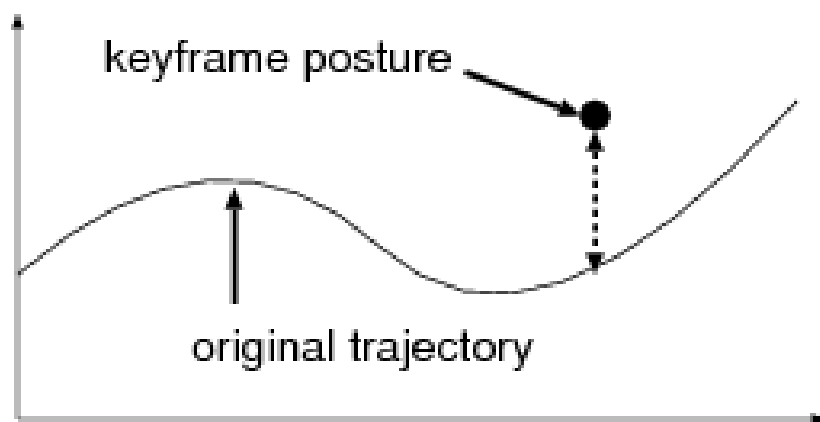
Motion Warping

- Edit the captured motion a little bit so that it satisfies the requirements
 - Effective for changing the location the hand or the foot passes



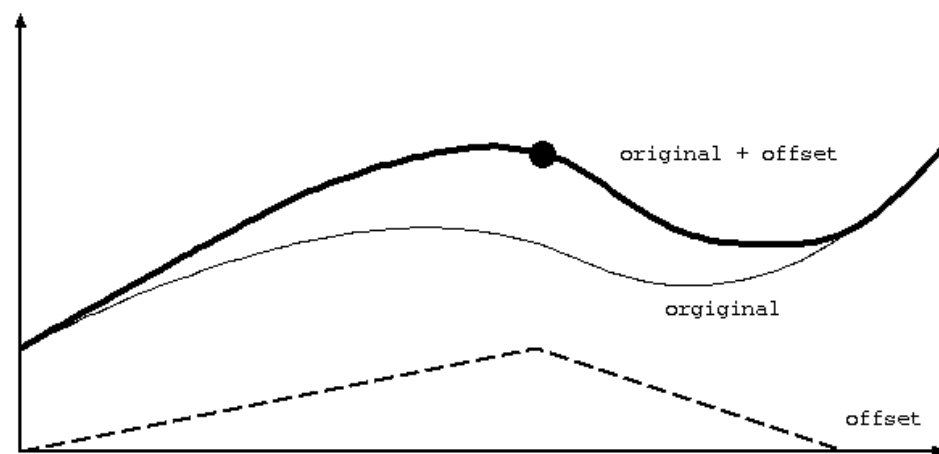
Basic idea

- Adding offset to the data so the constraint is satisfied



Warped Motion = original motion + offset

Offset can be a simple 1D motion

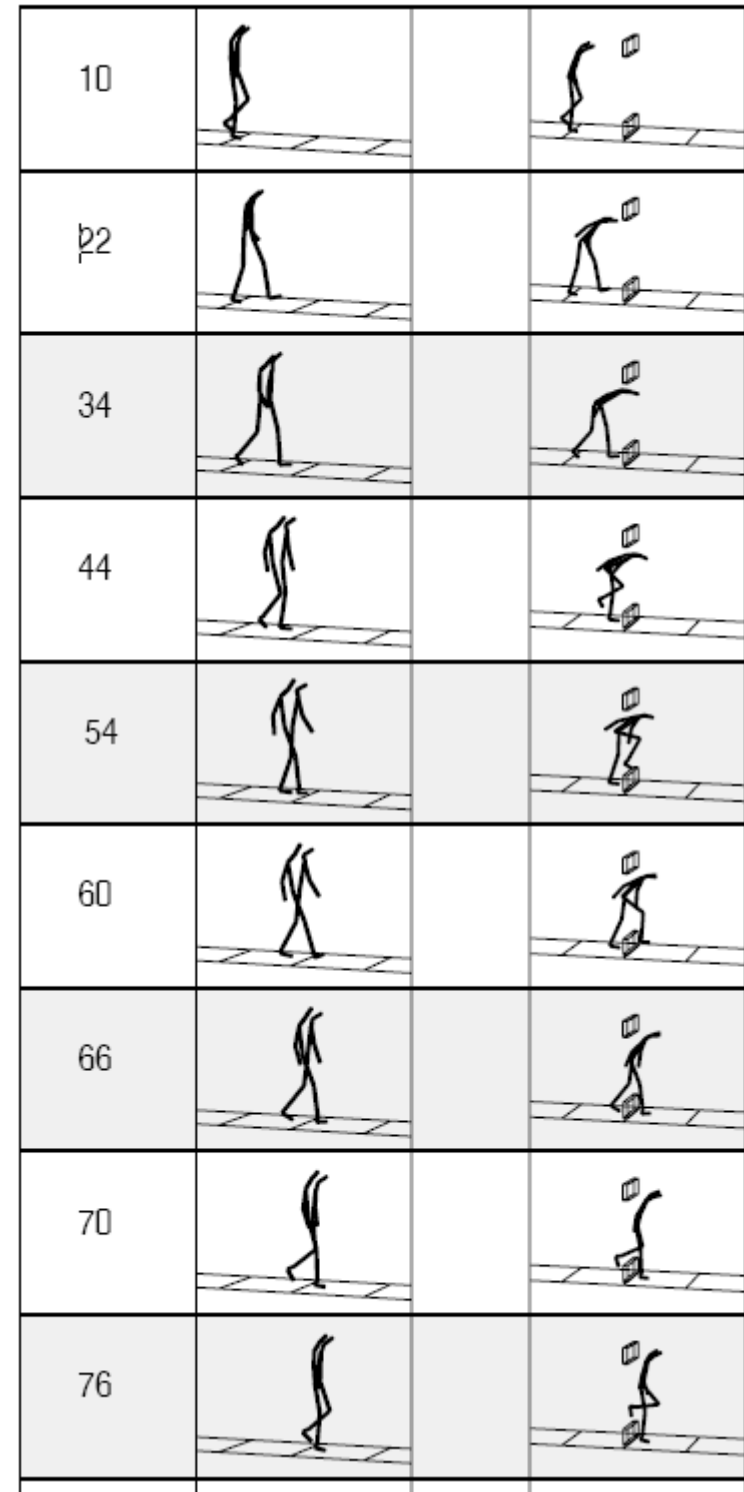


Motion Warping

- Very simple
- Good for avoiding obstacles

<http://www.youtube.com/watch?v=BzfxGwO7swg>

http://www.youtube.com/watch?v=Ro30uWsWI_s



**Tennis
Forehand**

Obstacles

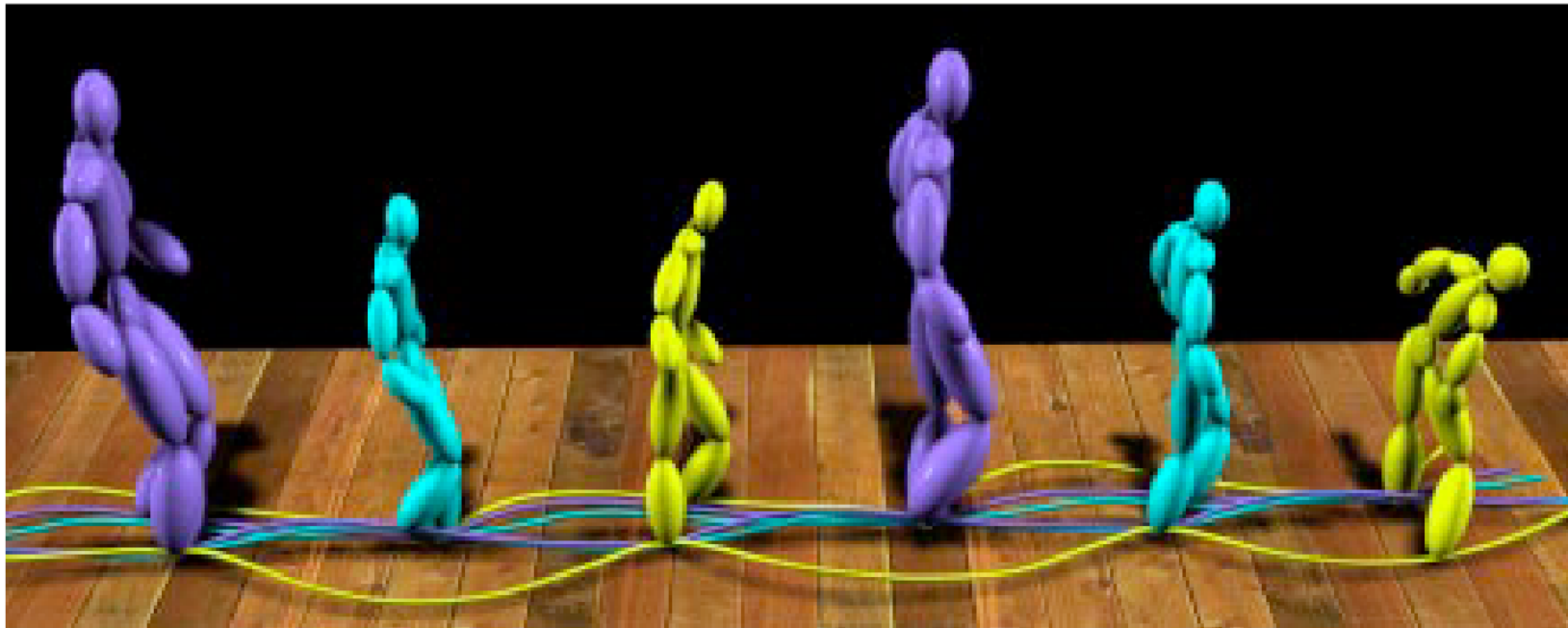
Motion Editing, Motion Retargeting

- Just adding offsets is not satisfactory in some cases
- For the dancing example, we need the hands to be connected during the entire motion
- Or the support foot must be on the ground all the time
- Then need to edit the motion by inverse kinematics instead of simply adding an offset to the motion



Motion Editing, Motion Retargeting

- Edit the end-effector trajectory
- Compute the pose of the character by inverse kinematics so that
 - It tracks the end effector positions
 - the motion is similar to the original motion



Motion Editing by Inverse Kinematics

Mapping from the joint angles $\mathbf{q}$ to the joint positions $\mathbf{x}$

$$\mathbf{x} = f(\mathbf{q})$$

We can linearize the equation and obtain the relation for the velocities:

$$\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}}$$

For computing the joint motion from end effector motion, we can use the pseudo inverse:

$$\dot{\mathbf{q}} = \mathbf{J}^+ \dot{\mathbf{x}}$$

$$\mathbf{J}^+ = \mathbf{J}^T (\mathbf{J}\mathbf{J}^T + \lambda^2 \mathbf{I})^{-1}$$

Tracking the end-effector movements

All solutions that satisfies the constraints:

$$\dot{\mathbf{q}} = \mathbf{J}^+ \dot{\mathbf{x}} + \underbrace{(\mathbf{I} - \mathbf{J}^+ \mathbf{J})}_{\text{null space term}} \mathbf{y}$$

This null space term corresponds to redundant degrees of freedom and can be used to perform secondary tasks.

For example, we can set $\mathbf{y}$ to the desired motion $\dot{\mathbf{q}}_{\text{des}}$ to make $\dot{\mathbf{q}}$ similar to $\dot{\mathbf{q}}_{\text{des}}$

$$\dot{\mathbf{q}} = \mathbf{J}^+ \dot{\mathbf{x}} + (\mathbf{I} - \mathbf{J}^+ \mathbf{J}) \dot{\mathbf{q}}_{\text{des}}$$

Tracking the end-effector movements

$$\dot{\mathbf{q}} = \mathbf{J}^+ \dot{\mathbf{x}} + \underbrace{(\mathbf{I} - \mathbf{J}^+ \mathbf{J}) \dot{\mathbf{q}}_{\text{des}}}_{\text{null space term}}$$

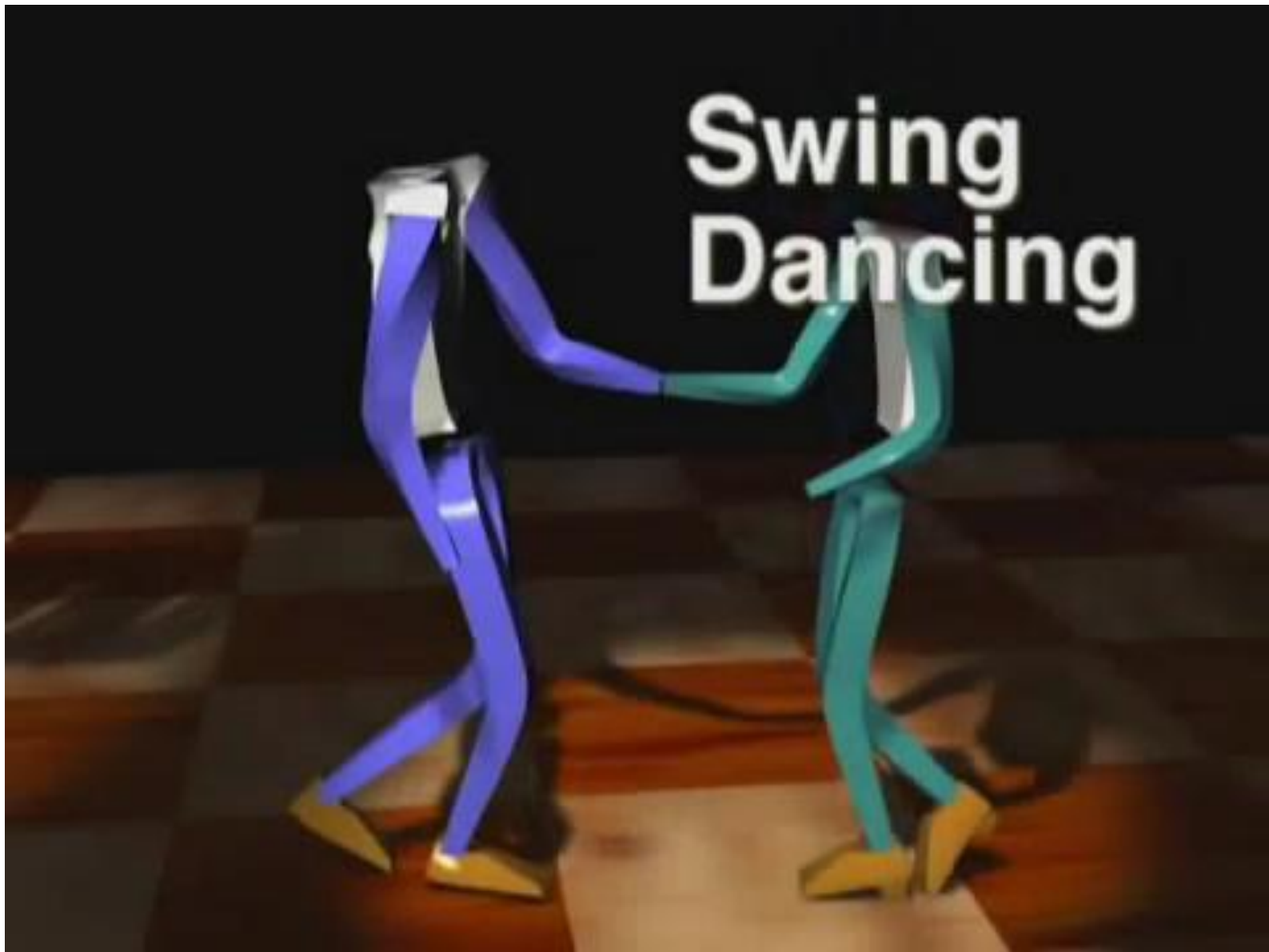
$(\mathbf{I} - \mathbf{J}^+ \mathbf{J}) \dot{\mathbf{q}}_{\text{des}}$ projects $\dot{\mathbf{q}}_{\text{des}}$ to a motion that does not move the end effector.

Remember $\mathbf{J}^+$ produces the minimum motion for the body to produce the end effector motion $\dot{\mathbf{x}}$:

$$\begin{aligned} (\mathbf{I} - \mathbf{J}^+ \mathbf{J}) \dot{\mathbf{q}}_{\text{des}} &= \dot{\mathbf{q}}_{\text{des}} - \mathbf{J}^+ \mathbf{J} \dot{\mathbf{q}}_{\text{des}} = \dot{\mathbf{q}}_{\text{des}} - \mathbf{J}^+ \dot{\mathbf{x}}_{\text{des}} \\ &= \underbrace{\dot{\mathbf{q}}_{\text{des}}}_{\text{this produces } \dot{\mathbf{x}}_{\text{des}}} - \underbrace{\hat{\dot{\mathbf{q}}}_{\text{des}}}_{\text{Minimal motion for producing } \dot{\mathbf{x}}_{\text{des}}} \end{aligned}$$

Subtracting the elements that produce the end effector motion

Swing Dancing



Editing/Blending Motion Styles

Editing the motion styles is a difficult problem but simple methods exists for locomotion

Editing/Blending the motion in the frequency domain is effective for stylizing the motion



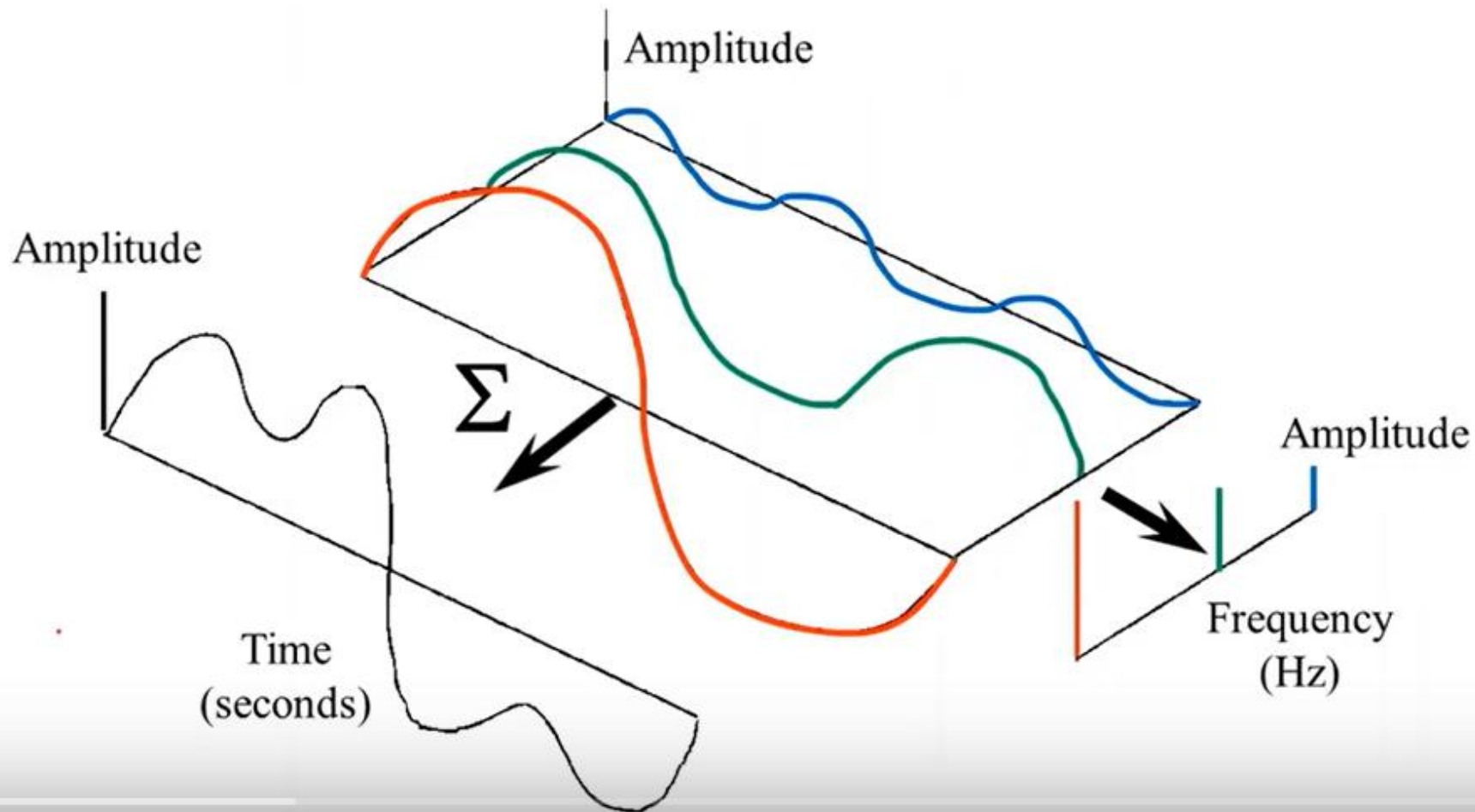
(a) Normal walk



(b) Tired walk

Fourier Transform

Any periodic signal can be expressed by a combination of a proper set of sine waves



Editing/Blending Motion Styles

We first do a Fourier analysis of each degrees of freedom of motion 1 (say normal walk)

$$q_m^A(t) = A_{m,0} + \sum_{n>0} A_{m,n} \sin(n t + \phi_{m,n})$$

Let's do this for another motion (say tired walk)

$$q_m^B(t) = B_{m,0} + \sum_{n>0} B_{m,n} \sin(n t + \psi_{m,n})$$

Editing/Blending Motion Styles

Then, we can interpolate/extrapolate the styles by interpolating the Fourier parameters

$$\begin{aligned} q_m^{A-B}(t) &= (1-s)A_{m,0} + sB_{m,0} \\ &+ \sum_{n>0} \left((1-s)A_{m,n} + sB_{m,n} \right) \sin(n t + \left((1-s)\phi_{m,n} + s\psi_{m,n} \right)) \end{aligned}$$



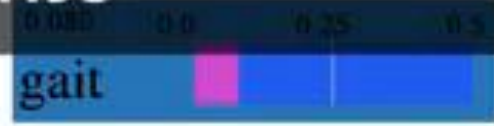
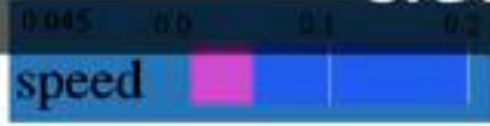
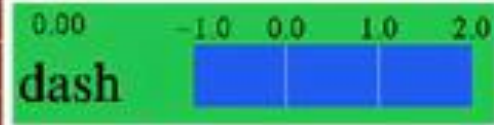
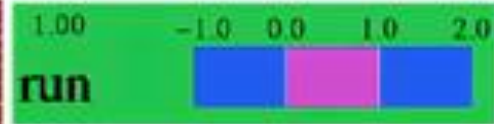
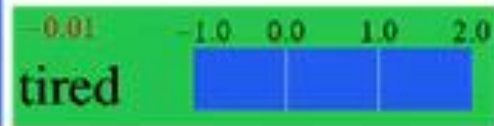
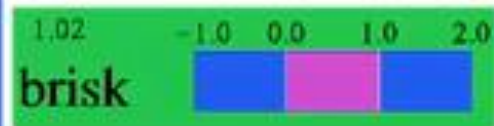
(c) $s = 0.5$



(d) $s = 2.0$



(e) $s = -0.5$



Fourier principles for emotion-based human figure animation

Munetoshi Unuma, Ken Anjyo, Ryoza Takeuchi

SIGGRAPH95

Motion Editing and Blending

Motion editing is good but if you edit too much, the motion can appear unnatural.

Probably it is better to prepare different motions and gradually switch between them to keep the motion natural

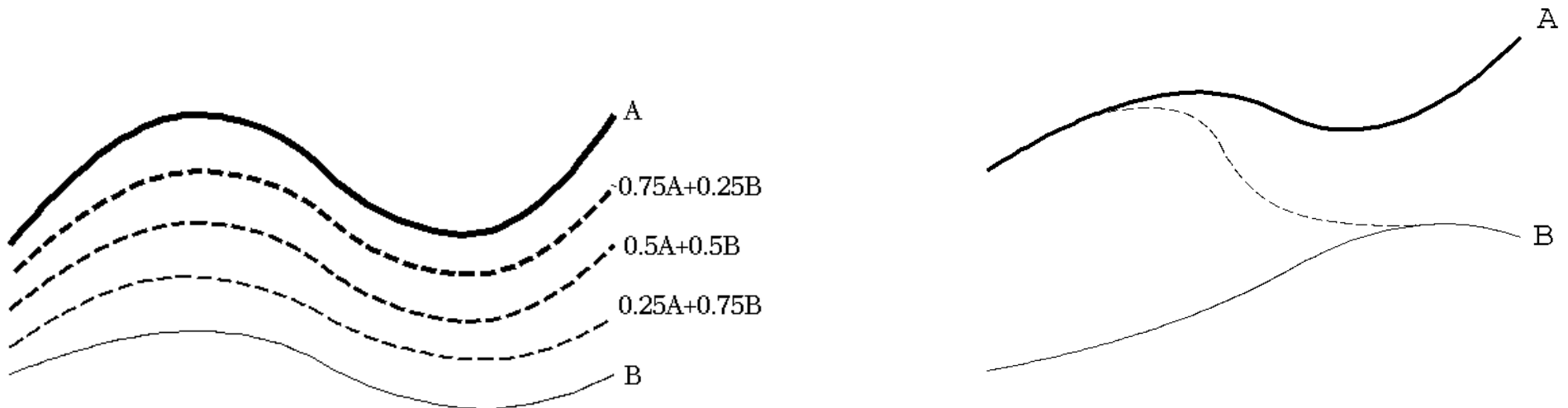
Motion Blending

- Given two different motions A and B

$$\text{Blended Motion} = (1 - w)A + wB \quad (0 < w < 1)$$

How can we use this?

- Generate a motion inbetween
- Gradually shift from motion A to motion B
- Concatenating two motions – blending the two ends by gradually shifting s from 0 to 1





Motion Blending: Blending Multiple Motions

Given example motion $M_1, \dots, M_n$

We can define parameters like

$(w_1, w_2, \dots, w_n)$ s.t.

$$w_1 + w_2 + \dots + w_n = 1$$

And then blend the motion by

$$w_1 M_1 + w_2 M_2 + \dots + w_n M_n$$



Motion Blending: Blending Multiple Motions

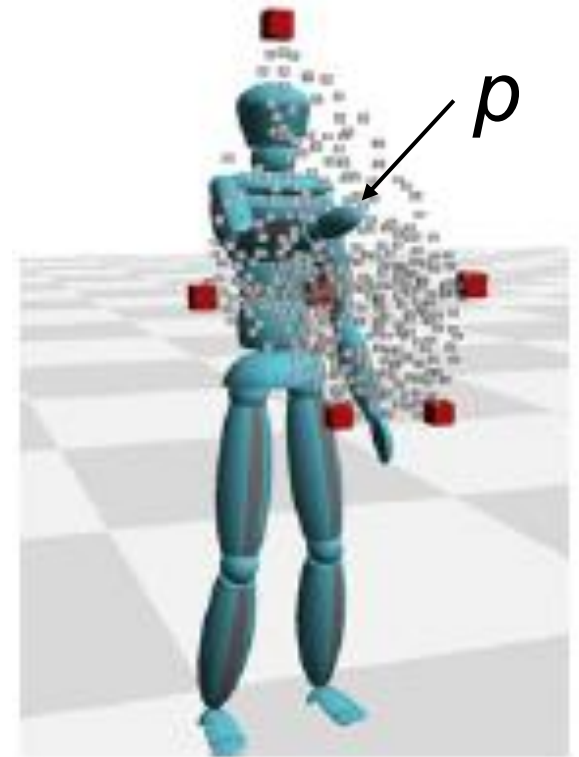
Computing the weights?

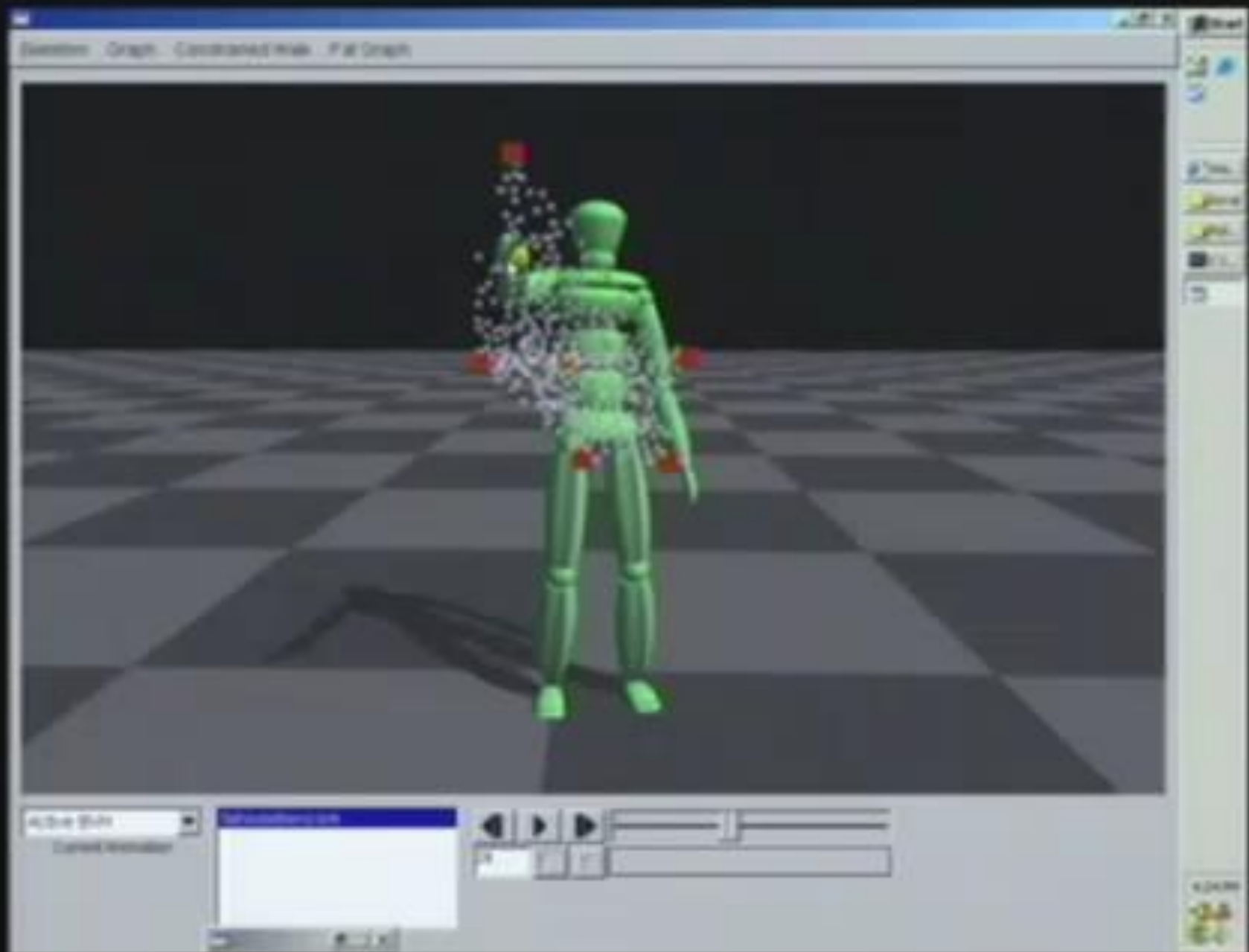
- Say we have a target hand position p :
- And the position for each motion is $(p_1, p_2, \dots, p_n)$,

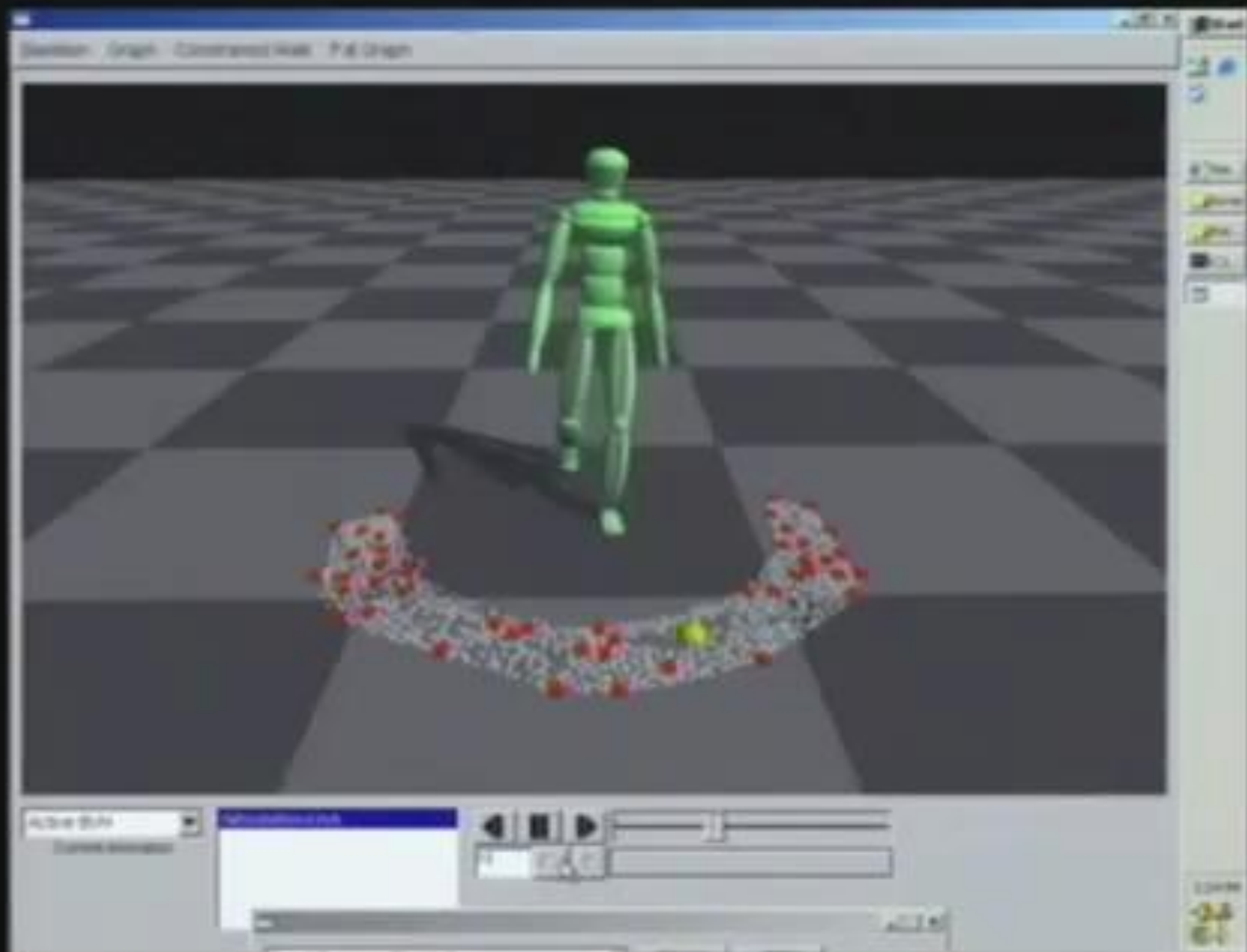
The weights can be computed by

$$w_i = a_i / (a_1 + a_2 + \dots + a_n)$$

$$\text{where } a_i = 1 / |p_i - p|$$

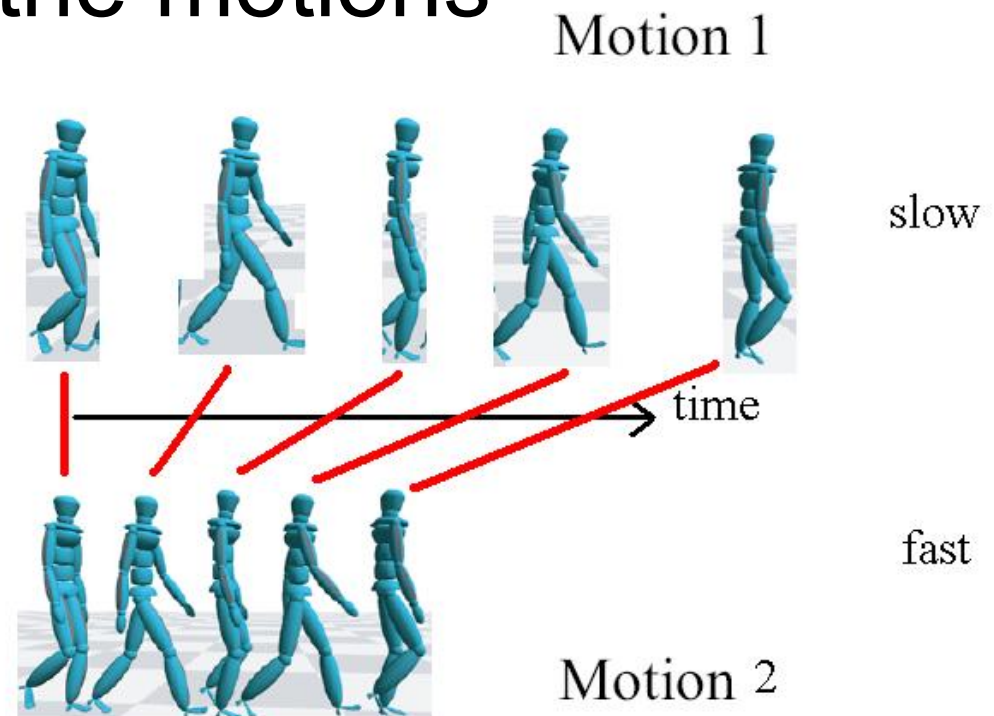






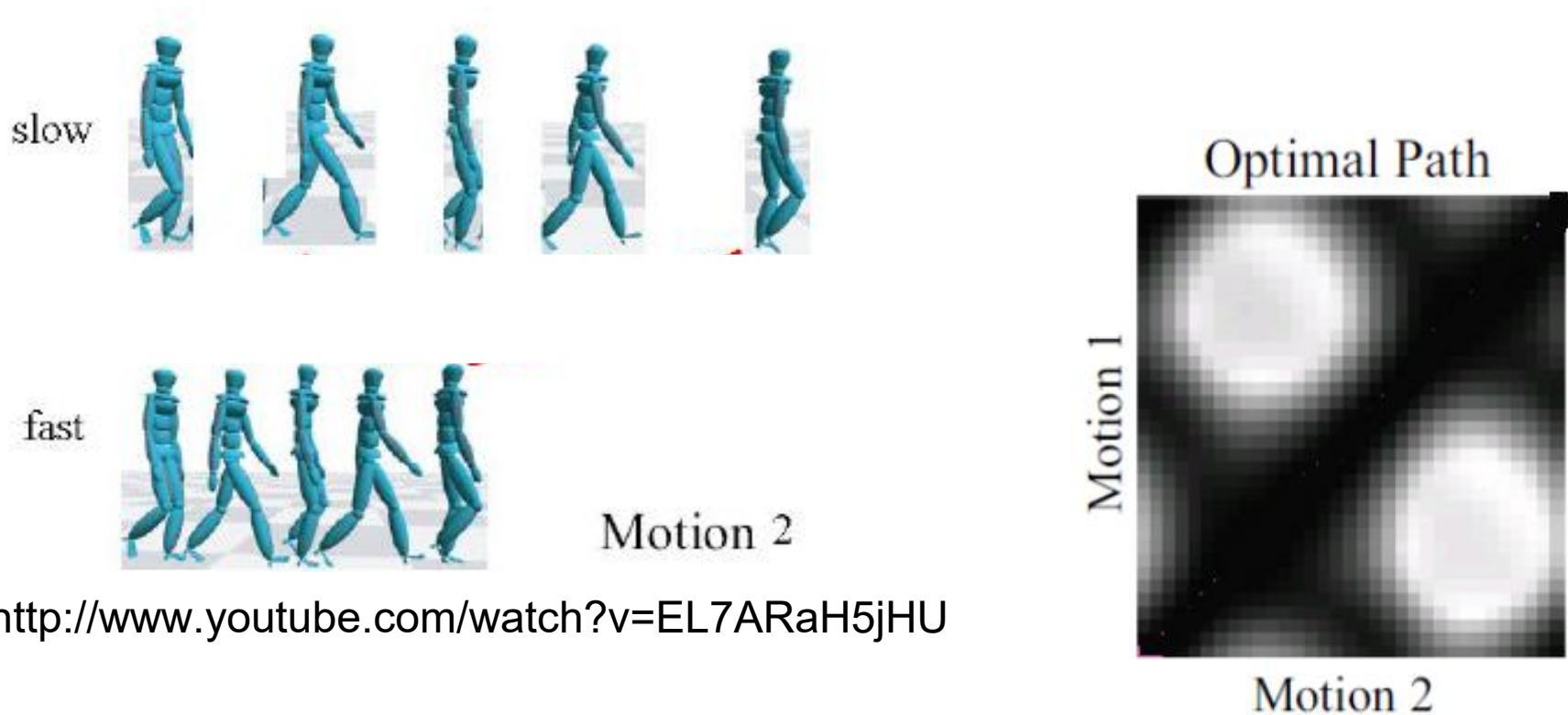
The problems when blending motions

- The durations of motions are different
- The timing of some events are different
 - Suppose we blend two walking motions
 - The timing of steps are different
 - Need to synchronize the motions



Dynamic Time Warping (DTW)

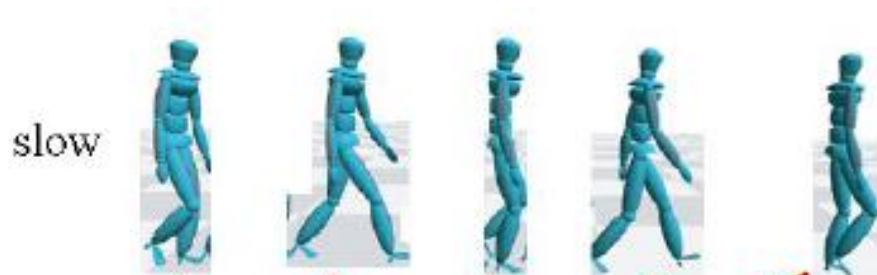
- Define the difference of postures at every frame in motion 1 and motion 2 : $D(\mathbf{q}_1, \mathbf{q}_2) = d(\mathbf{q}_1, \mathbf{q}_2) + \alpha d(\dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2)$
- Create a similarity matrix of all frames



<http://www.youtube.com/watch?v=EL7ARaH5jHU>

Dynamic Time Warping (DTW)

- Define the difference of postures at every frame in motion 1 and motion 2 :
- Create a similarity matrix of all frames
- Find the shortest path from the left bottom to the right top in this matrix (assume the matrix is a map) - this gives



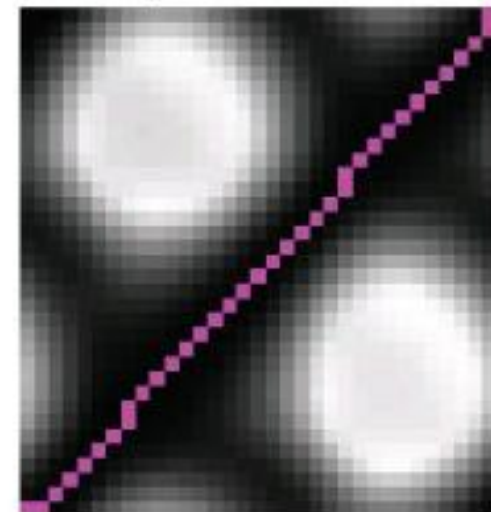
Motion 2

<http://www.youtube.com/watch?v=EL7ARaH5jHU>

$$\min \sum_i d_i(q_1, q_2)$$

Optimal Path

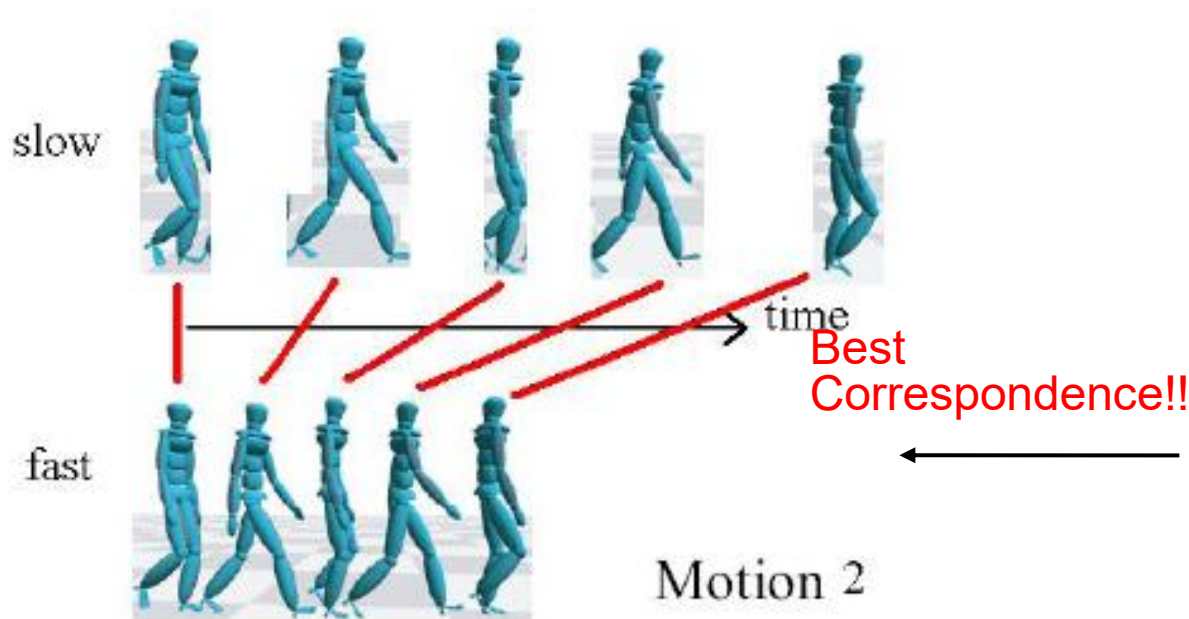
Motion 1



Motion 2

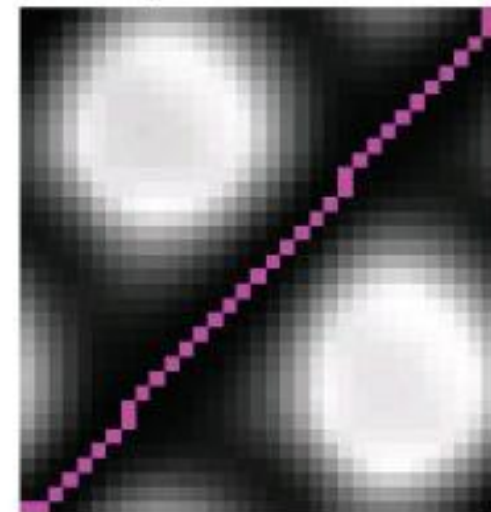
Dynamic Time Warping (DTW)

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Optimal Path



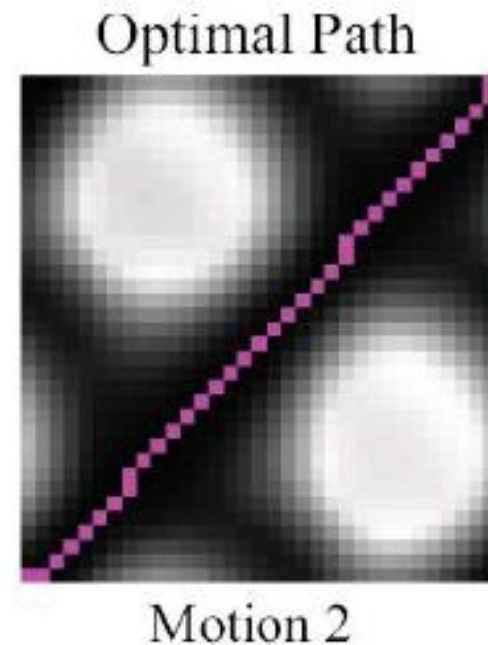
Motion 2

<http://www.youtube.com/watch?v=EL7ARaH5jHU>

Dynamic Time Warping (DTW)

- The computed distance also represents the similarity of the motion
- We can search for motion clips that are similar in the entire motion database

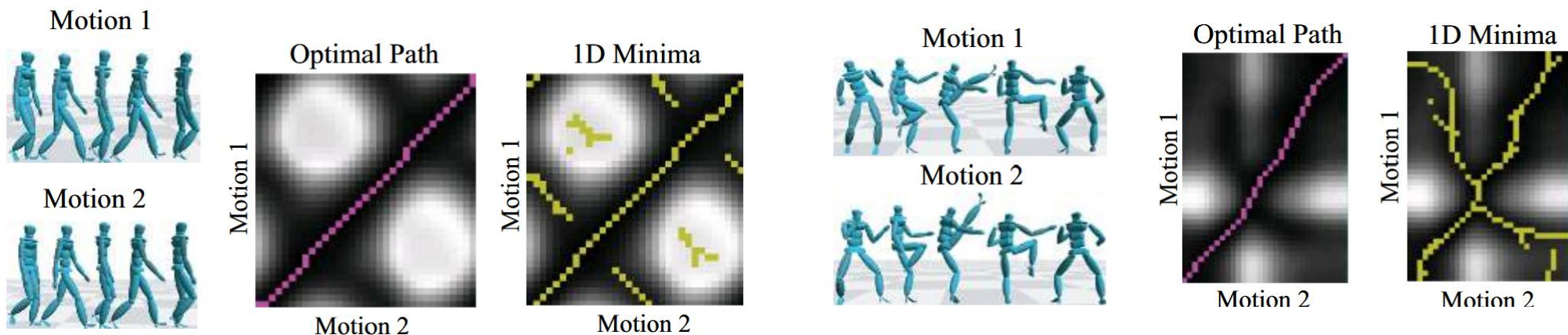
$$\min \sum_i d_i(q_1, q_2)$$



Searching Motion Data Sets

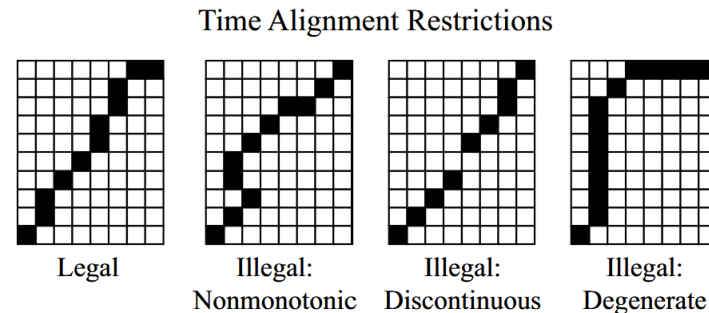
Finding similar motions in the dataset

- Using the idea of distance grid, we can find similar motions in the dataset
- We compute the distance between all the poses in the motion database and form a distance grid.
- We find the local minima in the grid (1D local minima in either horizontal or vertical direction)

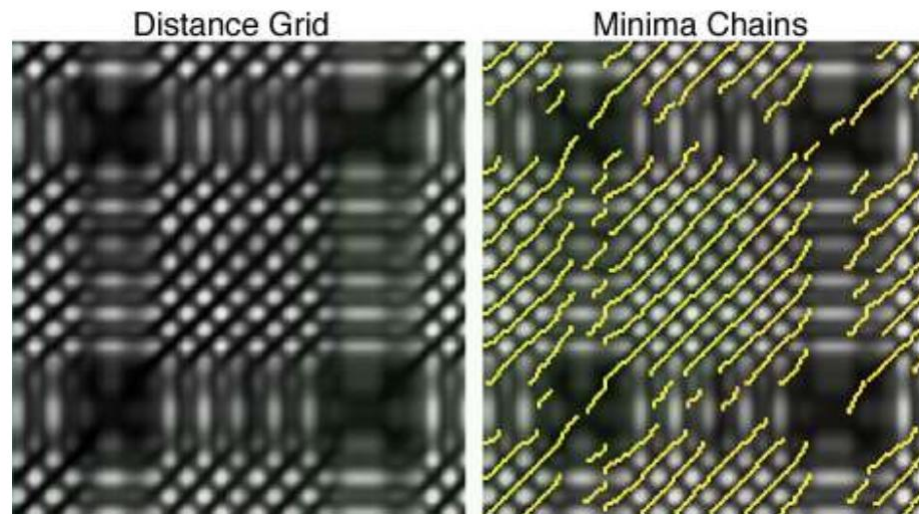


Finding similar motions in the dataset

- But we have rules that a valid transition is only along moving to the left/upward (time increasing)

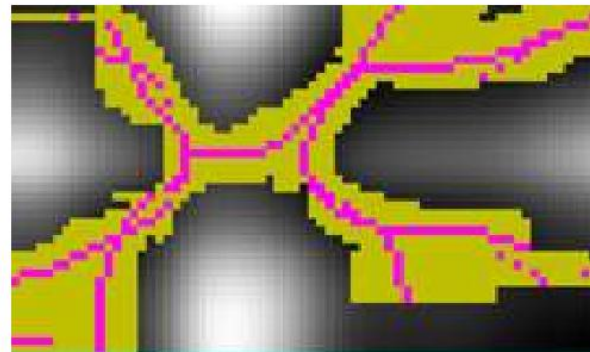


- We can then find the valid paths that can be used to find similar movements



Finding similar motions in the dataset

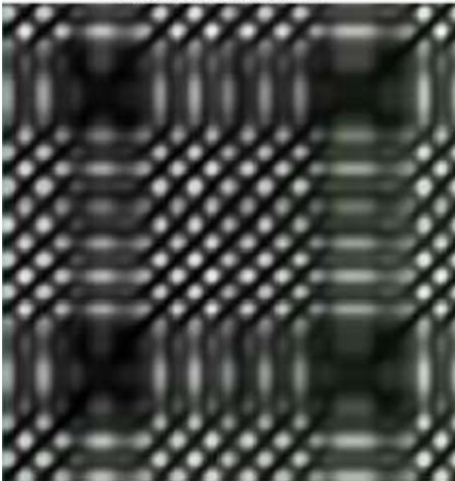
- The local minima might be a criterion that is too restrictive
- We can expand the frames to the adjacent ones within in threshold
- Then, find shortest paths within these regions to form longer movements



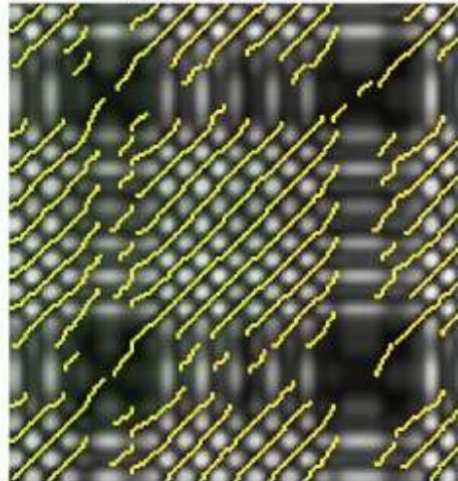
Local minima (magenta) are extended to form the valid region (yellow).

Finding similar motions in the dataset

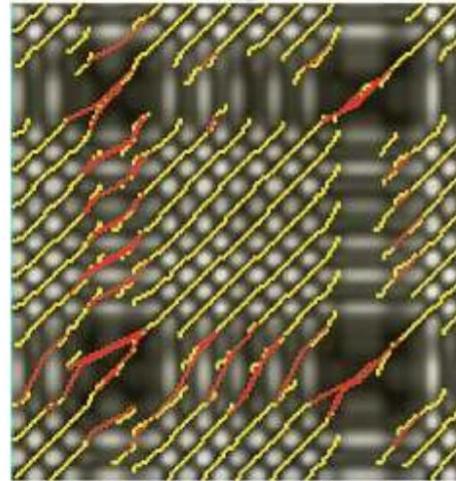
Distance Grid



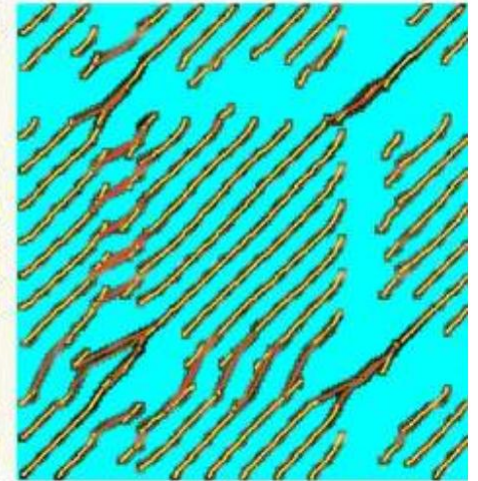
Minima Chains



Bridges



Refinement



Discussions: Motion Editing/Blending

- Motion editing: Editing a motion capture data to satisfy constraints, retargeting and change the style
- Motion blending: synthesize novel motion by blending different movements.
 - Can produce more natural motion
 - Data-driven motion synthesis

Motion Databases

Mixamo: <https://www.mixamo.com/#/>

Not only the motion but also the skin data

Website to easily download it

[https://github.com/ChrisWu1997/2D-Motion-](https://github.com/ChrisWu1997/2D-Motion-Retargeting/blob/master/dataset/Guide%20For%20Downloading%20Mixamo%20Data.md)

[Retargeting/blob/master/dataset/Guide%20For%20Downlo](https://github.com/ChrisWu1997/2D-Motion-Retargeting/blob/master/dataset/Guide%20For%20Downloading%20Mixamo%20Data.md)

[ing%20Mixamo%20Data.md](https://github.com/ChrisWu1997/2D-Motion-Retargeting/blob/master/dataset/Guide%20For%20Downloading%20Mixamo%20Data.md)

CMU : <http://mocap.cs.cmu.edu/>

- Various motion data in different topics
 - Locomotion
 - Physical activity and sports
 - Human interactions

Lafan1 : <https://github.com/ubisoft/ubisoft-laforge-animation-dataset>

Readings / References

- Witkin and Popovic “Motion Warping”, SIGGRAPH 95
- Choi, Kwang-Jin, and Hyeong-Seok Ko. "Online motion retargetting." *The Journal of Visualization and Computer Animation* 11.5 (2000): 223-235.
- Kovar and Gleicher Flexible Automatic Motion Blending with Registration Curves, SCA 2003
- Kovar and Gleicher, “Automated Extraction and Parameterization of Motions in Large Data Set”, SIGGRAPH 2004